Daily Inspection Digest

Sunday, 18 October 2026

1. Summary

Report ID	DIG-20261018
Date	2026-10-18
Inspection runs	2
Footage analysed	0m 27s
Confirmed events	5 total: 1 critical incidents, 3 in this digest, 1 inventory
Status	AI-detected, pending verification

Class	Events	Severity
Crack	1	Critical
Squat	1	Critical
Spalling	0	Critical
Flaking	1	Moderate
Groove	1	Moderate
Joint	1	Info

2. Escalations and Items Needing Review

Reference	Class	Confidence	Action	Video time
INC-20261018-0001	Crack	94%	Incident report issued	00:00.0
EVT-20261018-0004	Squat	30%	Low confidence: review	00:08.7

3. Findings Compiled in This Digest

Image	Event	Video time	Class	Severity	Conf.	Location
	EVT-20261018-0002	00:02.7	Flaking	Moderate	70%	12.84080, 80.15356 (sim.)
	EVT-20261018-0003	00:06.0	Groove	Moderate	60%	12.84105, 80.15376 (sim.)
	EVT-20261018-0004	00:08.7	Squat	Critical	30%	12.84125, 80.15392 (sim.)

4. Trend vs Previous Day

Class	Today	Previous day	Change
Crack	1	0	+1
Squat	1	0	+1
Spalling	0	0	+0
Flaking	1	0	+1
Groove	1	0	+1
Joint	1	0	+1

No events were recorded on the previous day.

5. Inspection Coverage

Run	Source	Model	Frames	Started
#1	single_crack.mp4	Faster R-CNN (ResNet-50 FPN)	200	18 Oct 2026, 17:49:27
#2	mixed_demo.mp4	Faster R-CNN (ResNet-50 FPN)	200	18 Oct 2026, 17:49:28

Inventory only: 1 rail joint/feature event(s) logged, not reported as defects.

6. Verification and Sign-off

Reviewer name	
Role / designation	
Decision	[ ] Confirmed defect [ ] False positive [ ] Needs re-inspection
Notes	
Signature	
Date	

Appendix: Model, Limitations and Disclaimer

Model	Faster R-CNN (ResNet-50 FPN)
Model benchmark	Project benchmark on the track-fault test set: mAP50 17.69%, precision 46.61%, recall 8.63%.
Weights file	fake_model.pth
Weights SHA-256 (first 16)	n/a
GUI confidence threshold	0.20
Event confirmation rule	3 detections within 6 frames; event closes after 15 frames unseen

Limitations

- Detections come from a machine-learning model and may include false positives or miss defects.
- Confidence is the model's score, not a calibrated probability.
- Sizes are in image pixels. No real-world scale is available unless the camera is calibrated.
- Location is simulated whenever the location source says so. Do not use it to dispatch crews.
- This report must be verified by a qualified person before any maintenance or operational decision.